

Unit-4 : Symbolic Reasoning Under Uncertainty

Unit-4 Topic Roadmap

Main Topic	Subtopics	What You Will Learn
Introduction to Uncertainty in AI	Need of uncertainty handling, incomplete knowledge, uncertain environments	Why AI systems cannot always use exact logic
Symbolic Reasoning	Symbolic representation, logical inference	How AI uses symbols and rules for reasoning
Reasoning Under Uncertainty	Probabilistic vs symbolic uncertainty	Different ways AI handles uncertain information
Nonmonotonic Reasoning	Definition, assumptions, changing conclusions	How AI changes decisions when new facts arrive
Monotonic vs Nonmonotonic Logic	Comparison, examples	Difference between classical logic and intelligent reasoning
Closed World Assumption (CWA)	Known vs unknown facts	How databases and AI systems assume missing facts
Default Reasoning	Default assumptions, exceptions	AI reasoning using common-sense assumptions
Circumscription	Minimizing assumptions	How AI limits unnecessary possibilities
Truth Maintenance System (TMS)	Dependency tracking, belief revision	Updating AI knowledge when facts change
Logics for Non-monotonic Reasoning	Default Logic, Autoepistemic Logic	Formal methods for uncertain reasoning
Applications	Expert systems, diagnosis systems, NLP, robotics	Real-world use of symbolic uncertain reasoning

Introduction to Uncertainty in Artificial Intelligence

1. What is Uncertainty?

Uncertainty means:

The AI system does not have complete, exact, or guaranteed information.

In real-world situations, AI often works with:

- Incomplete data
- Ambiguous information
- Changing environments
- Human errors
- Probabilities instead of exact facts

Because of this, AI systems cannot always make 100% certain decisions.

2. Why Uncertainty Exists in AI

Cause	Explanation
Incomplete Information	All facts are not available
Noisy Data	Sensors or inputs may contain errors
Changing Environment	Real-world situations change continuously
Human Language Ambiguity	Same sentence can have multiple meanings
Unknown Future Events	AI cannot predict everything perfectly

Conflicting Knowledge	Different rules may give different conclusions
------------------------------	--

3. Real-Life Examples of Uncertainty

Example	Uncertainty
Weather Prediction	Weather may change unexpectedly
Medical Diagnosis	Same symptoms may indicate different diseases
Self-driving Cars	Road conditions change dynamically
Voice Assistant	User speech may not be clear
Spam Detection	Email may or may not be spam

4. Traditional Logic vs Real-World AI

Traditional computer logic works like this:

Statement	Result
2 + 2 = 4	Always true
Water freezes at 0°C	Fixed rule

But real-world AI problems are not always exact.

Example:

- “Birds can fly.”

AI concludes:

- “Parrot can fly.”

Later new information arrives:

- “Parrot has injured wings.”

Now conclusion changes.

This creates uncertainty.

5. Need for Handling Uncertainty

AI systems must:

- Make decisions with limited information
- Handle changing knowledge
- Work with probabilities
- Revise conclusions when new facts arrive
- Mimic human reasoning

Without uncertainty handling, AI becomes rigid and unrealistic.

6. Types of Uncertainty in AI

Type	Meaning	Example
Data Uncertainty	Input data may be wrong	Faulty sensor reading
Knowledge Uncertainty	Knowledge base incomplete	Missing medical history
Prediction Uncertainty	Future unknown	Stock market prediction
Linguistic Uncertainty	Ambiguous language	“Bank” (river bank or money bank)
Randomness	Natural randomness	Dice game

7. Approaches to Handle Uncertainty

Method	Purpose
Probability Theory	Measures likelihood
Bayesian Networks	Probabilistic reasoning
Fuzzy Logic	Handles partial truth
Nonmonotonic Reasoning	Revises conclusions
Certainty Factors	Confidence-based reasoning
Dempster-Shafer Theory	Combines uncertain evidence

8. Symbolic Reasoning Under Uncertainty

In symbolic AI:

- Knowledge is represented using symbols and rules.
- AI reasons using logical statements.

But real-world logic is uncertain.

Therefore AI uses:

Symbolic Reasoning Under Uncertainty

This allows AI to:

- Make assumptions
- Handle exceptions
- Revise beliefs
- Work with incomplete knowledge

9. Monotonic vs Nonmonotonic Thinking

Monotonic Reasoning

Once a conclusion is true, it never changes.

Example:

- All humans are mortal.
- Ram is human.
- Therefore Ram is mortal.

New information does not change conclusion.

Nonmonotonic Reasoning

Conclusions can change when new information arrives.

Example:

- Birds fly.
- Penguin is a bird.
- Therefore Penguin flies.

New fact:

- Penguins cannot fly.

AI changes conclusion.

This is closer to human thinking.

10. Key Characteristics of Uncertain AI Systems

Characteristic	Description
Flexible	Can adapt to new information
Approximate	Gives near-correct answers
Probabilistic	Uses likelihood values
Dynamic	Handles changing environments
Intelligent	Mimics human reasoning

11. Applications of Uncertainty Handling

Application	Usage
Medical Expert Systems	Disease prediction
Robotics	Navigation decisions
NLP	Understanding ambiguous language
Autonomous Vehicles	Road decision making

Financial Systems	Risk analysis
Cybersecurity	Threat detection

12. Advantages

Advantage	Explanation
Realistic AI	Works in practical situations
Better Decision Making	Uses incomplete information effectively
Human-like Reasoning	Can revise decisions
Adaptive Systems	Handles changing environments

13. Limitations

Limitation	Explanation
Complex Calculations	Probability computations may be large
Difficult Knowledge Representation	Uncertain facts are harder to store
Computational Cost	Requires more processing
Not Always Accurate	Predictions may still fail

14. Important Terms

Term	Meaning
Uncertainty	Lack of complete certainty
Inference	Deriving conclusions
Knowledge Base	Collection of rules and facts
Belief Revision	Updating conclusions
Probability	Chance of occurrence
Symbolic AI	AI using symbols and logic

15. Short Exam Definition

Uncertainty in AI

Uncertainty in Artificial Intelligence refers to situations where the AI system does not possess complete or exact information, causing decisions and conclusions to be probabilistic or changeable.

16. One-Line Viva Answers

Question	Answer
Why uncertainty occurs in AI?	Because real-world information is incomplete and dynamic.
Which AI methods handle uncertainty?	Probability, fuzzy logic, Bayesian networks, and nonmonotonic reasoning.
What is uncertain reasoning?	Reasoning with incomplete or changing knowledge.
Which reasoning mimics human thinking?	Nonmonotonic reasoning.

Symbolic Reasoning Basics

1. Introduction to Symbolic Reasoning

Symbolic Reasoning is a method in Artificial Intelligence where knowledge is represented using:

- Symbols
- Facts

- Rules
- Logical statements

The AI system uses these symbols and logical rules to draw conclusions and solve problems.

2. Meaning of “Symbolic”

A symbol represents an object, action, relation, or idea.

Examples:

Real-World Object	Symbol
Human	HUMAN(X)
Bird	BIRD(X)
Flying ability	CAN_FLY(X)
Parent relation	PARENT(X,Y)

Instead of storing raw data only, AI stores meaning using symbols.

3. Definition

Symbolic Reasoning is the process of representing knowledge using symbols and applying logical rules to derive conclusions.

4. Need for Symbolic Reasoning

AI systems require symbolic reasoning because:

Need	Explanation
Knowledge Representation	Stores facts in understandable form
Logical Decision Making	Helps AI think systematically
Problem Solving	Used in planning and reasoning
Human-Like Intelligence	Mimics logical human thinking
Explainability	AI can explain why a conclusion was reached

5. Components of Symbolic Reasoning

Component	Description
Symbols	Represent objects or concepts
Facts	Statements known to be true
Rules	Conditions used for reasoning
Inference Engine	Applies logical rules
Knowledge Base	Collection of facts and rules

6. Knowledge Representation in Symbolic AI

Knowledge is represented in structured forms such as:

Method	Description
Predicate Logic	Uses predicates and variables
Production Rules	IF-THEN rules
Semantic Networks	Graph-based relationships
Frames	Structured object representation

7. Example of Symbolic Reasoning

Facts

- Bird(Tweety)
- Bird(X) $\rightarrow$ CanFly(X)

Reasoning

AI checks:

- Tweety is a bird.
- All birds can fly.

Conclusion

- CanFly(Tweety)

This process is called inference.

8. Inference in Symbolic Reasoning

Inference means deriving new information from existing facts and rules.

General Process

1. Store facts
2. Apply rules
3. Generate conclusions

9. Types of Inference

Type	Description
Deductive Inference	General to specific
Inductive Inference	Specific to general
Abductive Inference	Best possible explanation

10. Deductive Reasoning

In deductive reasoning:

- Rules are always logically valid.
- If premises are true, conclusion is definitely true.

Example

- All humans are mortal.
- Ram is human.

Conclusion:

- Ram is mortal.

11. Inductive Reasoning

In inductive reasoning:

- AI observes patterns.
- Conclusions are probable, not guaranteed.

Example

- Sun rose every day before.
- Therefore sun will rise tomorrow.

This may be true, but not absolutely certain.

12. Abductive Reasoning

Abductive reasoning finds the most likely explanation.

Example

- Patient has fever and cough.
- AI predicts flu as likely disease.

Used heavily in:

- Medical diagnosis
- Fault detection
- Expert systems

13. Rule-Based Reasoning

Rules are commonly represented as:

IF condition THEN action

Example

IF:

- Person has fever

THEN:

- Possible infection

These rules are stored in the knowledge base.

14. Production Rules

Production systems use condition-action rules.

Structure

IF condition → THEN conclusion

Example

IF:

- Bird(X)

THEN:

- CanFly(X)

15. Knowledge Base

The knowledge base stores:

- Facts
- Rules
- Relationships

It acts as the memory of the AI system.

16. Inference Engine

The inference engine:

- Reads facts
- Applies rules
- Produces conclusions

It is called the “brain” of the reasoning system.

17. Forward and Backward Reasoning

Method	Description
Forward Reasoning	Starts from facts toward conclusion
Backward Reasoning	Starts from goal and checks facts

18. Forward Reasoning Example

Facts

- Fever
- Cough

Rule

IF fever AND cough → Flu

Conclusion

- Flu detected

19. Backward Reasoning Example

Goal

Check whether patient has flu.

AI checks:

- Does patient have fever?
- Does patient have cough?

If yes:

- Goal satisfied.

20. Characteristics of Symbolic Reasoning

Characteristic	Explanation
Logical	Uses formal logic
Explainable	Provides reasoning steps
Knowledge Driven	Depends on stored knowledge
Structured	Organized representation
Rule-Based	Uses IF-THEN rules

21. Advantages of Symbolic Reasoning

Advantage	Description
Easy Interpretation	Human-readable reasoning
Explainability	AI decisions can be traced
Good for Expert Systems	Useful in rule-based applications
Strong Logical Support	Accurate logical reasoning

22. Limitations of Symbolic Reasoning

Limitation	Description
Difficult Knowledge Creation	Large rule bases required
Poor Learning Ability	Cannot learn automatically easily
Handles Uncertainty Poorly	Classical symbolic logic expects exact facts
Scalability Issues	Complex systems become large

23. Applications of Symbolic Reasoning

Application	Usage
Expert Systems	Medical and technical diagnosis
Natural Language Processing	Language understanding
Robotics	Decision making
Automated Theorem Proving	Mathematical proofs
Planning Systems	Goal-based actions

24. Symbolic AI vs Modern AI

Symbolic AI	Modern Data-Driven AI
Rule-based	Data-based
Explainable	Often black-box
Uses logic	Uses statistical learning
Human knowledge required	Learns from data
Best for reasoning	Best for pattern recognition

25. Relationship with Uncertainty

Traditional symbolic reasoning assumes:

- Knowledge is exact
- Rules are fixed

But real-world problems contain uncertainty.

Therefore advanced symbolic reasoning includes:

- Nonmonotonic reasoning
- Fuzzy logic
- Probabilistic reasoning

These methods improve decision making under uncertainty.

Nonmonotonic Reasoning

1. Introduction

Traditional logic assumes that:

Once a conclusion becomes true, it always remains true.

But real-world situations change continuously.

New information may force the AI system to revise its previous conclusions.

This type of reasoning is called:

Nonmonotonic Reasoning

2. Definition

Nonmonotonic Reasoning is a type of logical reasoning in which conclusions can be withdrawn or modified when new information becomes available.

It allows AI systems to:

- Make assumptions
- Handle incomplete knowledge
- Revise beliefs dynamically

3. Why Nonmonotonic Reasoning is Needed

Real-world AI systems often work with:

- Incomplete information
- Uncertain environments
- Exceptions to rules
- Changing facts

Classical logic cannot handle these situations effectively.

4. Example of Nonmonotonic Reasoning

Initial Knowledge

- Birds can fly.
- Penguin is a bird.

Initial Conclusion

- Penguin can fly.

New Information

- Penguins cannot fly.

Updated Conclusion

- Penguin cannot fly.

The previous conclusion is removed after receiving new knowledge.

This is nonmonotonic behavior.

5. Meaning of “Nonmonotonic”

In mathematics and logic:

- “Monotonic” means conclusions only increase and never decrease.
- “Nonmonotonic” means conclusions may change or be removed.

6. Characteristics of Nonmonotonic Reasoning

Characteristic	Description
Dynamic	Conclusions may change
Flexible	Handles changing knowledge
Human-Like	Mimics common-sense reasoning
Assumption-Based	Uses temporary assumptions
Exception Handling	Supports exceptions to rules

7. Monotonic vs Nonmonotonic Reasoning

Feature	Monotonic Reasoning	Nonmonotonic Reasoning
Conclusion Change	Never changes	Can change
Knowledge Nature	Fixed	Dynamic
Exception Handling	Poor	Good
Real-World Suitability	Limited	High
Human-Like Thinking	Less	More

8. Working Principle

Nonmonotonic reasoning follows this process:

1. AI receives partial information.
2. AI makes temporary assumptions.
3. AI draws conclusions.
4. New information arrives.
5. AI revises old conclusions if necessary.

9. Common-Sense Reasoning

Humans naturally use nonmonotonic reasoning.

Example

- Normally, students attend college.
- Ravi is a student.
- Therefore Ravi attends college.

New fact:

- Ravi is on medical leave.

Conclusion changes:

- Ravi may not attend college now.

10. Default Reasoning

Default reasoning is an important part of nonmonotonic reasoning.

AI assumes something is true unless evidence proves otherwise.

Example

- Birds normally fly.
- Sparrow is a bird.

AI assumes:

- Sparrow flies.

This is called a default assumption.

11. Closed World Assumption (CWA)

In many AI systems:

If something is not known to be true, it is assumed to be false.

This is called Closed World Assumption.

Example

Database contains:

- Student(Amit)

But no information about:

- Student(Riya)

AI assumes:

- Riya is not a student.

12. Open World vs Closed World

Feature	Open World	Closed World
Missing Information	Unknown	Assumed false
Knowledge	Incomplete	Treated as complete
Used In	Web systems	Databases

13. Belief Revision

Belief revision means updating knowledge when new information arrives.

AI systems must:

- Remove invalid conclusions
- Add new valid conclusions
- Maintain consistency

14. Truth Maintenance System (TMS)

A Truth Maintenance System helps AI:

- Track dependencies between beliefs
- Remove incorrect conclusions
- Maintain logical consistency

15. Example of Belief Revision

Initial Facts

- Roads are clear.
- Car can move fast.

New Fact

- Heavy traffic detected.

Revised Conclusion

- Car should move slowly.

16. Importance in Artificial Intelligence

Nonmonotonic reasoning is important because real-world AI systems require:

- Flexible thinking
- Adaptive decision-making
- Exception handling
- Human-like intelligence

17. Applications of Nonmonotonic Reasoning

Application	Usage
Expert Systems	Revising diagnosis
Robotics	Dynamic environment decisions
NLP	Understanding changing context
Autonomous Vehicles	Real-time decision revision
Planning Systems	Updating plans dynamically

18. Advantages

Advantage	Explanation
Handles Incomplete Knowledge	Works without full information
Supports Exceptions	Manages special cases
Human-Like Reasoning	Similar to practical thinking
Flexible Decision Making	Adapts to new knowledge

19. Limitations

Limitation	Explanation
Complex Implementation	Difficult to design
High Computational Cost	Frequent updates required
Knowledge Management Difficulty	Large systems become complex
Possibility of Contradictions	Conflicting conclusions may occur

20. Types of Nonmonotonic Reasoning

Type	Description
Default Reasoning	Uses default assumptions
Circumscription	Minimizes assumptions
Autoepistemic Logic	Reasoning about own beliefs
Belief Revision	Updating conclusions dynamically

21. Nonmonotonic Logic

Nonmonotonic logic is the formal logical system used to represent:

- Changing conclusions
- Assumptions
- Exceptions
- Incomplete knowledge

It extends classical logic to support intelligent reasoning.

22. Real-World Example in AI

Medical Diagnosis System

Initial Symptoms

- Fever
- Headache

AI predicts:

- Viral infection

New Test Report

- Malaria parasite detected

AI revises conclusion:

- Malaria

This dynamic updating is nonmonotonic reasoning.

23. Importance in Modern AI

Modern AI systems such as:

- Chatbots
- Self-driving cars
- Smart assistants
- Diagnostic systems

must continuously revise decisions according to new information.

Therefore nonmonotonic reasoning is essential for practical AI.

24. Relationship with Symbolic AI

Nonmonotonic reasoning is part of symbolic AI because:

- Knowledge is represented symbolically.
- Logical rules are used.
- Conclusions are updated dynamically.

It combines:

- Logic
- Knowledge representation
- Intelligent decision-making

Monotonic vs Nonmonotonic Logic

1. Introduction

Logic in Artificial Intelligence is used to derive conclusions from known facts and rules.

Two major types of logical reasoning are:

1. Monotonic Logic
2. Nonmonotonic Logic

These differ mainly in how conclusions behave when new information is added.

2. Monotonic Logic

Definition

Monotonic Logic is a type of reasoning in which previously derived conclusions never change after adding new information.

Once a conclusion becomes true:

- it remains true forever.

3. Working of Monotonic Logic

In monotonic logic:

- Knowledge continuously increases.
- Conclusions only increase.
- No conclusion is removed.

4. Example of Monotonic Logic

Facts

- All humans are mortal.
- Ram is a human.

Conclusion

- Ram is mortal.

Even if new information is added later, this conclusion remains valid.

5. Characteristics of Monotonic Logic

Characteristic	Description
Stable Conclusions	Conclusions never change
Fixed Knowledge	Assumes complete information
Strict Logical Rules	Uses classical logic
No Exception Handling	Cannot manage changing facts
Predictable	Behavior always consistent

6. Advantages of Monotonic Logic

Advantage	Explanation
Simple Reasoning	Easy to implement
Mathematically Sound	Strong logical foundation
Consistent Results	No belief revision required
Efficient for Fixed Domains	Works well in static systems

7. Limitations of Monotonic Logic

Limitation	Explanation
Cannot Handle Exceptions	Real-world rules often have exceptions
No Belief Revision	Conclusions cannot change
Poor Real-World Adaptability	Real environments are dynamic
Assumes Complete Knowledge	Often unrealistic

8. Nonmonotonic Logic

Definition: Nonmonotonic Logic is a type of reasoning in which conclusions may change when new information becomes available.

The AI system can:

- Withdraw old conclusions
- Revise beliefs
- Handle exceptions dynamically

9. Working of Nonmonotonic Logic

In nonmonotonic logic:

1. AI uses available knowledge.
2. AI makes assumptions.
3. AI derives conclusions.
4. New information may invalidate old conclusions.

10. Example of Nonmonotonic Logic

Initial Facts

- Birds can fly.
- Penguin is a bird.

Initial Conclusion

- Penguin can fly.

New Information

- Penguins cannot fly.

Revised Conclusion

- Penguin cannot fly.

The earlier conclusion is removed.

11. Characteristics of Nonmonotonic Logic

Characteristic	Description
Dynamic Conclusions	Conclusions may change
Flexible Reasoning	Adapts to new facts
Exception Handling	Supports unusual cases
Human-Like Thinking	Similar to common-sense reasoning
Assumption-Based	Works with incomplete knowledge

12. Advantages of Nonmonotonic Logic

Advantage	Explanation
Handles Real-World Problems	Suitable for dynamic systems
Supports Incomplete Knowledge	Works even with missing information
Flexible Decision Making	Adapts to changes
Better Human Reasoning Model	Mimics practical thinking

13. Limitations of Nonmonotonic Logic

Limitation	Explanation
Complex Implementation	Difficult system design
Higher Computational Cost	Frequent updates required
Contradiction Handling Needed	Conflicts may occur
Large Knowledge Management	Complex for huge systems

14. Comparison Between Monotonic and Nonmonotonic Logic

Feature	Monotonic Logic	Nonmonotonic Logic
Conclusion Behavior	Never changes	Can change
Knowledge Nature	Static	Dynamic
New Information Effect	Only adds conclusions	May remove conclusions
Exception Handling	Not supported	Supported
Real-World Suitability	Limited	High
Human-Like Reasoning	Less natural	More natural
Assumptions	Rarely used	Frequently used
Belief Revision	Not possible	Possible
Flexibility	Low	High
Complexity	Simple	Complex

15. Mathematical Interpretation

Monotonic Logic

If:

$$A \subseteq B$$

then:

$$\text{Conclusions}(A) \subseteq \text{Conclusions}(B)$$

Adding more knowledge never reduces conclusions.

Nonmonotonic Logic

Adding new knowledge may:

- change conclusions
- remove earlier conclusions

Therefore monotonic property is not preserved.

16. Real-Life Comparison

Situation	Monotonic Logic	Nonmonotonic Logic
Weather Prediction	Fixed forecast	Forecast updated dynamically
Medical Diagnosis	Single final diagnosis	Diagnosis revised after tests
Self-driving Cars	Fixed decisions	Decisions updated continuously
Human Thinking	Rigid	Flexible

17. Applications of Monotonic Logic

Application	Usage
Mathematical Proof Systems	Fixed logical rules
Database Queries	Stable information
Formal Verification	Software correctness
Theorem Proving	Exact reasoning

18. Applications of Nonmonotonic Logic

Application	Usage
Expert Systems	Revising diagnosis
Robotics	Dynamic decision making
NLP	Context understanding
Intelligent Agents	Adaptive reasoning
Autonomous Systems	Real-time belief updates

19. Importance in AI

Modern AI systems operate in environments where:

- Information changes continuously
- Exceptions occur frequently
- Knowledge is incomplete

Therefore:

- Classical monotonic logic alone is insufficient.
- Nonmonotonic logic becomes essential.

20. Relationship with Human Thinking

Human reasoning is naturally nonmonotonic.

Humans:

- make assumptions
- revise decisions
- adapt to new evidence

AI systems use nonmonotonic logic to imitate this behavior.

21. Example Using Student Attendance

Monotonic Logic

- All registered students attend classes.
- Ravi is registered.

Conclusion:

- Ravi attends classes.

This conclusion never changes.

Nonmonotonic Logic

Initial Rule

- Students normally attend classes.

Initial Conclusion

- Ravi attends classes.

New Information

- Ravi is hospitalized.

Revised Conclusion

- Ravi may not attend classes.

22. Role in Symbolic Reasoning

Both monotonic and nonmonotonic logic belong to symbolic AI because they use:

- Symbols
- Logical rules
- Inference mechanisms

But nonmonotonic logic provides:

- flexibility
- uncertainty handling
- practical intelligence

for real-world reasoning.

Closed World Assumption (CWA)

1. Introduction

In Artificial Intelligence and database systems, knowledge is often incomplete.

To simplify reasoning, many systems use an assumption called:

Closed World Assumption (CWA)

This assumption helps AI decide what to do when information is missing.

2. Definition

Closed World Assumption states that if a fact is not known to be true, then it is assumed to be false.

In simple words:

- Known facts → True
- Unknown facts → False

3. Basic Idea

Under CWA:

“What is not stored in the knowledge base does not exist.”

The AI system assumes that the knowledge base contains all necessary information.

4. Example of Closed World Assumption

Database Facts

- Student(Amit)
- Student(Rahul)

No information about:

- Student(Riya)

AI Conclusion

- Amit is a student → True
- Rahul is a student → True
- Riya is a student → False

Because Riya is not present in the database.

5. Working Principle of CWA

1. AI checks the knowledge base.
2. If a fact exists → True
3. If a fact does not exist → Assume False

6. Characteristics of Closed World Assumption

Characteristic	Description
Assumption-Based	Missing facts treated as false
Database-Oriented	Commonly used in databases
Simplifies Reasoning	Easier decision making
Efficient	Fast querying and inference
Nonmonotonic Nature	New facts can change conclusions

7. Why CWA is Nonmonotonic

Under CWA:

- AI may assume something is false initially.
- Later new information may prove it true.

Example

Initially:

- No record of Employee(Ravi)

AI assumes:

- Ravi is not an employee.

Later:

- Employee(Ravi) added.

Now conclusion changes.

Therefore CWA supports nonmonotonic reasoning.

8. Open World Assumption vs Closed World Assumption

Feature	Closed World Assumption	Open World Assumption
Missing Information	Assumed false	Treated as unknown
Knowledge Base	Considered complete	Considered incomplete
Common Usage	Databases	Semantic web, ontology systems
Decision Style	Strict	Flexible

9. Example Comparison

Closed World

Database:

- Car(Toyota)

No record of:

- Car(Honda)

Conclusion:

- Honda is not a car.

Open World

No information about Honda means:

- Unknown whether Honda is a car.

10. Applications of CWA

Application	Usage
Relational Databases	SQL query systems
Expert Systems	Rule-based assumptions
Information Retrieval	Missing data handling
Logic Programming	Prolog systems
Access Control Systems	Permission checking

11. Advantages of CWA

Advantage	Explanation
Simple Reasoning	Easy implementation
Faster Decisions	Missing facts immediately treated false
Efficient Database Queries	Reduced ambiguity
Practical for Limited Domains	Works well when data is mostly complete

12. Limitations of CWA

Limitation	Explanation
Incorrect Assumptions Possible	Missing data may actually exist
Not Suitable for Dynamic Knowledge	Real-world information changes frequently
Incomplete Knowledge Problems	Unknown facts may be wrongly rejected
Limited Real-World Flexibility	Too rigid for uncertain environments

13. Real-World Example

Library System

Database contains:

- Borrowed(Book1)
- Borrowed(Book2)

No record of:

- Borrowed(Book3)

System assumes:

- Book3 is available.

This is Closed World Assumption.

14. Relationship with Nonmonotonic Reasoning

CWA is part of nonmonotonic reasoning because:

- Conclusions are assumption-based.
- Conclusions may change when new facts are added.
- AI revises beliefs dynamically.

Default Reasoning

1. Introduction

Humans often make decisions using normal assumptions unless exceptions are found.

Artificial Intelligence uses a similar concept called:

Default Reasoning

It allows AI systems to make reasonable conclusions even when complete information is unavailable.

2. Definition

Default Reasoning is a type of reasoning in which AI assumes something to be true by default unless evidence proves otherwise.

It is based on:

- common sense
- typical situations
- default assumptions

3. Basic Idea

Default reasoning works as:

“Normally true unless an exception exists.”

4. Example of Default Reasoning

Rule

- Birds normally fly.

Fact

- Sparrow is a bird.

Default Conclusion

- Sparrow can fly.

Exception

New fact:

- Sparrow has broken wings.

Updated conclusion:

- Sparrow cannot fly.

5. Structure of Default Reasoning

General form:

IF condition AND no contradiction

THEN assume conclusion

6. Characteristics of Default Reasoning

Characteristic	Description
Assumption-Based	Uses default assumptions
Handles Incomplete Knowledge	Works without full information
Supports Exceptions	Conclusions can change
Human-Like Reasoning	Similar to practical thinking
Nonmonotonic Nature	Conclusions are revisable

7. Why Default Reasoning is Needed

Real-world systems rarely have complete information.

Without default reasoning:

- AI cannot make decisions quickly.
- Systems become overly dependent on exact facts.

Default reasoning improves:

- flexibility

- practical reasoning
- intelligent behavior

8. Example Using Students

Default Rule

- Students normally attend classes.

Fact

- Ravi is a student.

Conclusion

- Ravi attends classes.

Exception

New information:

- Ravi is hospitalized.

Updated conclusion:

- Ravi may not attend classes.

9. Default Assumptions

A default assumption is a temporary belief accepted until contradiction appears.

Examples

Default Assumption	Possible Exception
Birds fly	Penguins do not fly
Cars have four wheels	Some special vehicles differ
Employees come to office	Work-from-home cases

10. Working Process of Default Reasoning

1. AI stores default rules.
2. AI checks available facts.
3. AI applies default assumptions.
4. AI derives conclusions.
5. AI revises conclusions if exceptions appear.

11. Applications of Default Reasoning

Application	Usage
Expert Systems	Medical assumptions
Robotics	Environment assumptions
NLP	Understanding incomplete language
Autonomous Vehicles	Predicting normal behavior
Intelligent Agents	Decision making

12. Advantages of Default Reasoning

Advantage	Explanation
Fast Decision Making	Works with incomplete information
Human-Like Thinking	Mimics common-sense reasoning
Flexible	Handles changing knowledge
Practical for Real World	Supports uncertainty

13. Limitations of Default Reasoning

Limitation	Explanation
Incorrect Assumptions Possible	Defaults may be wrong
Exception Handling Complexity	Many exceptions increase difficulty
Contradictions May Occur	Conflicting defaults possible
Computational Complexity	Large systems become difficult

14. Default Logic

Default reasoning is formally represented using:

Default Logic

Default logic extends classical logic by:

- allowing assumptions
- handling exceptions
- supporting incomplete knowledge

15. Relationship with Nonmonotonic Reasoning

Default reasoning is a major form of nonmonotonic reasoning because:

- conclusions are temporary
- assumptions may change
- new information can invalidate conclusions

16. Real-World Example

Medical Diagnosis

Default Rule

- Fever usually indicates infection.

Initial Conclusion

- Patient may have infection.

New Lab Report

- Fever caused by allergy.

Revised Conclusion

- Infection diagnosis removed.

This is default reasoning with belief revision.

Circumscription

1. Introduction

In real-world reasoning, humans usually assume that:

If something is not known to be abnormal, then it is considered normal.

Artificial Intelligence uses a similar concept called:

Circumscription

Circumscription is a nonmonotonic reasoning technique used to minimize assumptions and handle incomplete knowledge.

2. Definition

Circumscription is a form of nonmonotonic reasoning in which AI assumes that only known facts are true and anything not proven abnormal is considered normal.

It minimizes:

- exceptions
- abnormal situations
- unnecessary assumptions

3. Basic Idea of Circumscription

Circumscription works on the principle:

“Assume normal conditions unless evidence shows otherwise.”

AI tries to:

- reduce unknown abnormalities
- simplify reasoning
- make practical conclusions

4. Example of Circumscription

General Rule

- Birds fly.

Known Exception

- Penguins cannot fly.

If AI knows:

- Sparrow is a bird.

And no abnormal condition is mentioned:

AI assumes:

- Sparrow can fly.

This happens because AI minimizes abnormal cases.

5. Why Circumscription is Needed

Real-world AI systems cannot store every possible normal condition explicitly.

Circumscription helps AI:

- make default assumptions
- reduce complexity
- handle incomplete knowledge efficiently

6. Working Principle

Circumscription follows these steps:

1. Store normal rules.
2. Identify known exceptions.
3. Minimize abnormal cases.
4. Assume everything else behaves normally.

7. Concept of Abnormality

Circumscription commonly uses an abnormality predicate.

Example

- $\text{Bird}(X) \wedge \text{NOT Abnormal}(X) \rightarrow \text{CanFly}(X)$

Meaning:

- If X is a bird
- and X is not abnormal
- then X can fly

8. Example with Abnormality

Facts

- $\text{Bird}(\text{Tweety})$
- $\text{Penguin}(\text{Polo})$
- $\text{Penguin}(X) \rightarrow \text{Abnormal}(X)$

Rule

- Bird(X) AND NOT Abnormal(X) $\rightarrow$ CanFly(X)

Conclusions

- Tweety can fly
- Polo cannot fly

Because Polo is marked abnormal.

9. Characteristics of Circumscription

Characteristic	Description
Nonmonotonic	Conclusions can change
Assumption-Based	Assumes normal situations
Exception Handling	Handles abnormal cases
Knowledge Minimization	Reduces unnecessary assumptions
Logical Reasoning	Uses formal symbolic logic

10. Circumscription as Nonmonotonic Reasoning

Circumscription is nonmonotonic because:

- AI initially assumes normal behavior.
- New abnormal information may change conclusions.

Example

Initial:

- Bird(Rio)

AI concludes:

- Rio can fly.

Later:

- Injured(Rio)

AI updates:

- Rio cannot fly.

The earlier conclusion is revised.

11. Types of Circumscription

Type	Description
Predicate Circumscription	Minimizes predicates
Formula Circumscription	Minimizes formulas
Prioritized Circumscription	Some assumptions given higher priority

12. Predicate Circumscription

Most common form.

AI minimizes abnormal predicates.

Example

- Minimize Abnormal(X)

Meaning:

- Assume as few abnormal objects as possible.

13. Relationship with Default Reasoning

Circumscription and default reasoning are closely related.

Circumscription	Default Reasoning
-----------------	-------------------

Minimizes abnormalities	Uses default assumptions
Formal logical method	Practical reasoning method
Logic-based	Rule-based
Assumes normality	Assumes common situations

Both support:

- incomplete knowledge
- exceptions
- flexible reasoning

14. Applications of Circumscription

Application	Usage
Expert Systems	Default medical assumptions
Robotics	Normal environment assumptions
NLP	Common-sense understanding
Knowledge Representation	Managing exceptions
Automated Planning	Assuming normal conditions

15. Advantages of Circumscription

Advantage	Explanation
Handles Incomplete Knowledge	Useful when full data unavailable
Supports Common-Sense Reasoning	Mimics human assumptions
Reduces Complexity	Minimizes unnecessary cases
Good Exception Handling	Easily manages abnormal situations

16. Limitations of Circumscription

Limitation	Explanation
Computational Complexity	Difficult for large systems
Hard Exception Modeling	Complex abnormal situations
Knowledge Engineering Difficulty	Requires proper rule design
Contradiction Possibility	Multiple assumptions may conflict

17. Real-World Example

Traffic System

Default Assumption

- Roads are open.

AI Conclusion

- Vehicle can travel normally.

New Information

- Road under construction.

Revised Conclusion

- Vehicle must take another route.

AI minimized abnormal conditions initially.

18. Circumscription in Knowledge Representation

Circumscription helps represent:

- defaults
- exceptions
- normal behaviors

- incomplete information

It improves intelligent reasoning in symbolic AI systems.

19. Importance in Artificial Intelligence

Circumscription is important because:

- Real-world information is incomplete.
- AI must assume normal conditions.
- Exception handling is essential.
- Human reasoning naturally minimizes abnormalities.

It provides:

- practical intelligence
- adaptive reasoning
- common-sense behavior

20. Relationship with Symbolic AI

Circumscription belongs to symbolic AI because it uses:

- logical formulas
- predicates
- symbolic representations
- inference mechanisms

It extends classical logic to support:

- uncertainty
- assumptions
- dynamic reasoning

Truth Maintenance System (TMS)

1. Introduction

Artificial Intelligence systems continuously receive new information.

Sometimes:

- old conclusions become invalid
- assumptions change
- contradictions appear

To handle these situations, AI uses:

Truth Maintenance System (TMS)

A TMS helps maintain consistency in the knowledge base by updating beliefs and conclusions dynamically.

2. Definition

A Truth Maintenance System (TMS) is a mechanism in Artificial Intelligence that keeps track of beliefs, assumptions, dependencies, and justifications, and updates conclusions whenever knowledge changes.

3. Main Goal of TMS

The primary goal of TMS is:

To maintain logically consistent knowledge in an AI system.

4. Why TMS is Needed

In intelligent systems:

- knowledge changes frequently

- assumptions may become false
- conclusions may conflict

Without TMS:

- AI may store incorrect beliefs
- contradictions may remain unresolved
- reasoning becomes unreliable

5. Basic Idea of TMS

TMS works by:

1. Storing facts and assumptions
2. Recording dependencies between beliefs
3. Tracking justifications for conclusions
4. Revising conclusions when facts change

6. Important Terms in TMS

Term	Meaning
Belief	Information accepted as true
Assumption	Temporary accepted fact
Justification	Reason for a conclusion
Dependency	Relationship between facts
Contradiction	Conflict between beliefs
Belief Revision	Updating conclusions

7. Components of TMS

Component	Description
Knowledge Base	Stores facts and rules
Assumption Set	Stores temporary assumptions
Justification Network	Tracks reasoning relationships
Dependency Manager	Maintains belief dependencies
Contradiction Handler	Detects inconsistencies

8. Working Principle of TMS

Step-by-Step Process

1. AI receives facts and assumptions.
2. AI derives conclusions.
3. TMS records why each conclusion was made.
4. New information arrives.
5. If contradiction occurs:
 - invalid conclusions are removed
 - affected beliefs are updated
6. Knowledge base becomes consistent again.

9. Example of TMS

Initial Facts

- Bird(Tweety)
- Birds can fly

Conclusion

- Tweety can fly

TMS stores:

- why the conclusion was made
- which facts support it

New Information

- Tweety is injured

Now TMS:

- removes old conclusion
- updates reasoning

New Conclusion

- Tweety cannot fly

10. Dependency Network

TMS creates a dependency graph showing:

- which conclusions depend on which facts
- how beliefs are connected

Example

Fact/Rule	Depends On
CanFly(Tweety)	Bird(Tweety), BirdsFly
Bird(Tweety)	Database fact

If one supporting fact changes:

- dependent conclusions also change.

11. Justification in TMS

A justification explains:

Why a conclusion is believed to be true.

Example

Conclusion:

- Patient has malaria

Justification:

- Fever
- Blood test positive
- Medical rules

12. Contradiction Handling

TMS detects contradictions automatically.

Example

Belief 1

- Road is open

New Information

- Road is closed

Contradiction detected.

TMS:

- removes invalid belief
- updates route planning

13. Belief Revision

Belief revision is one of the most important functions of TMS.

It includes:

- deleting invalid conclusions
- adding updated beliefs
- maintaining consistency

14. Types of Truth Maintenance Systems

Type	Description
JTMS	Justification-based Truth Maintenance System
ATMS	Assumption-based Truth Maintenance System
LTMS	Logic-based Truth Maintenance System

15. Justification-Based TMS (JTMS)

In JTMS:

- each belief has a justification
- system tracks dependency relationships

Features

Feature	Description
Tracks reasoning	Stores conclusion sources
Detects contradictions	Updates invalid beliefs
Simple structure	Easier implementation

16. Assumption-Based TMS (ATMS)

ATMS manages:

- multiple assumptions
- alternative belief sets

It is useful in:

- uncertain reasoning
- planning systems
- hypothesis testing

Features

Feature	Description
Multiple assumptions	Supports many possibilities
Efficient dependency tracking	Handles complex reasoning
Parallel reasoning	Multiple solutions possible

17. Logic-Based TMS (LTMS)

LTMS uses:

- propositional logic
- logical consistency checking

It is stronger than JTMS for formal reasoning.

18. Characteristics of TMS

Characteristic	Description
Dynamic	Updates beliefs continuously

Nonmonotonic	Conclusions may change
Consistency Maintenance	Removes contradictions
Dependency Tracking	Tracks reasoning paths
Intelligent Revision	Revises invalid conclusions

19. Relationship with Nonmonotonic Reasoning

TMS is closely related to nonmonotonic reasoning because:

- conclusions are revisable
- assumptions may change
- old beliefs can be removed

TMS provides practical implementation for:

- belief revision
- assumption handling
- dynamic reasoning

20. Applications of TMS

Application	Usage
Expert Systems	Revising diagnosis
Robotics	Updating environment knowledge
Planning Systems	Revising plans dynamically
NLP	Managing context changes
Autonomous Vehicles	Real-time decision updates
Intelligent Agents	Adaptive reasoning

21. Real-World Example

Medical Diagnosis System

Initial Symptoms

- Fever
- Headache

AI concludes:

- Viral infection

New Test Result

- Malaria positive

TMS:

- removes old diagnosis
- updates knowledge

Revised Conclusion

- Malaria

22. Advantages of TMS

Advantage	Explanation
Maintains Consistency	Removes contradictory beliefs
Supports Dynamic Knowledge	Handles changing information
Improves Reliability	More accurate reasoning
Tracks Justifications	Explains conclusions
Supports Nonmonotonic Reasoning	Flexible decision making

23. Limitations of TMS

Limitation	Explanation
Complex Implementation	Difficult system design
High Memory Usage	Dependency storage required
Computational Cost	Frequent updates expensive
Large Knowledge Management	Hard for massive systems

24. TMS vs Traditional Knowledge Systems

Feature	Traditional System	TMS
Belief Revision	Not supported	Supported
Contradiction Handling	Weak	Strong
Dependency Tracking	Minimal	Extensive
Dynamic Knowledge	Limited	Excellent
Real-World Adaptability	Low	High

25. Importance in Artificial Intelligence

TMS is important because modern AI systems require:

- adaptive reasoning
- belief management
- dynamic knowledge updates
- contradiction resolution

Without TMS:

- intelligent reasoning becomes unreliable in changing environments.

26. Summary of TMS Workflow

Initial Phase

- Store facts and assumptions

Reasoning Phase

- Generate conclusions

Tracking Phase

- Record justifications and dependencies

Update Phase

- Detect contradictions
- Revise beliefs
- Maintain consistency

27. Relationship with Symbolic AI

TMS belongs to symbolic AI because it uses:

- symbolic facts
- logical rules
- dependency graphs
- inference systems

It strengthens symbolic reasoning by enabling:

- dynamic belief management
- intelligent updating
- uncertainty handling

Autoepistemic Logic

1. Introduction

In Artificial Intelligence, systems sometimes need to reason not only about facts but also about:

- what they know
- what they do not know
- what they believe

This type of reasoning is called:

Autoepistemic Logic

The word:

- “Auto” → self
- “Epistemic” → knowledge or belief

So, Autoepistemic Logic means:

Logic for reasoning about one’s own knowledge and beliefs.

2. Definition

Autoepistemic Logic is a nonmonotonic reasoning system in which an AI system reasons about its own beliefs and knowledge.

It allows AI to:

- make assumptions
- analyze missing information
- revise beliefs dynamically

3. Basic Idea

In classical logic:

- reasoning is based only on facts.

In autoepistemic logic:

- reasoning also depends on what the system knows or does not know.

4. Example of Autoepistemic Reasoning Knowledge

- AI does not know that road is closed.

AI assumes:

- road is open.

New Information

- Road closure detected.

AI updates belief:

- road is closed.

This self-belief handling is autoepistemic reasoning.

5. Main Concept

Autoepistemic logic uses statements like:

“I believe X”

or

“I do not know X”

This enables self-aware reasoning.

6. Knowledge Operator

Autoepistemic logic commonly uses:

L(X)

Meaning:

- "X is believed to be true."

Example

L(CarRunning)

Meaning:

- AI believes the car is running.

7. Negation in Autoepistemic Logic

Expression:

NOT L(X)

means:

- X is not believed to be true.

Example

NOT L(Rain)

Meaning:

- AI does not believe it is raining.

8. Characteristics of Autoepistemic Logic

Characteristic	Description
Self-Reasoning	AI reasons about its own beliefs
Nonmonotonic	Conclusions may change
Belief-Based	Uses belief states
Dynamic	Handles changing knowledge
Assumption-Oriented	Makes conclusions from incomplete knowledge

9. Working Principle

Step-by-Step

1. AI stores knowledge.
2. AI analyzes known beliefs.
3. AI identifies unknown information.
4. AI makes assumptions.
5. New information may revise beliefs.

10. Example Using Weather System

Initial Knowledge

- AI has no information about rain.

AI assumes:

- Weather is clear.

New Sensor Data

- Rain detected.

AI revises:

- Weather is rainy.

11. Relationship with Nonmonotonic Reasoning

Autoepistemic logic is a type of nonmonotonic reasoning because:

- conclusions depend on beliefs
- assumptions may change
- new information revises old conclusions

12. Difference Between Classical Logic and Autoepistemic Logic

Feature	Classical Logic	Autoepistemic Logic
Uses Facts Only	Yes	No
Reasons About Beliefs	No	Yes
Dynamic Conclusions	Limited	Strong
Handles Incomplete Knowledge	Weak	Good
Self-Knowledge	Not supported	Supported

13. Applications of Autoepistemic Logic

Application	Usage
Intelligent Agents	Self-decision making
Robotics	Environment belief management
Expert Systems	Belief revision
Planning Systems	Reasoning with unknown information
Autonomous Vehicles	Dynamic environment assumptions

14. Advantages of Autoepistemic Logic

Advantage	Explanation
Supports Self-Reasoning	AI analyzes its own beliefs
Handles Incomplete Knowledge	Makes assumptions intelligently
Flexible Reasoning	Adapts to new information
Useful for Intelligent Agents	Supports autonomous thinking

15. Limitations of Autoepistemic Logic

Limitation	Explanation
Complex Implementation	Difficult formal logic
Computationally Expensive	Large belief systems costly
Contradiction Management	Conflicting beliefs possible
Hard Knowledge Representation	Requires advanced modeling

16. Relationship with Default Reasoning

Autoepistemic logic is closely related to default reasoning.

Default Reasoning	Autoepistemic Logic
Assumes normal conditions	Assumes based on beliefs
Uses defaults	Uses self-knowledge
Handles exceptions	Handles unknown beliefs

17. Importance in AI

Modern AI systems need:

- self-awareness
- belief management
- uncertainty handling
- adaptive intelligence

Autoepistemic logic provides these capabilities.

18. Real-World Example

Smart Home AI

Initial Situation

- No motion detected.

AI believes:

- Nobody is home.

New Information

- Motion sensor activated.

AI updates belief:

- Someone is inside the house.

This belief revision uses autoepistemic reasoning.

Applications and Case Studies of Symbolic Reasoning Under Uncertainty

1. Introduction

Symbolic reasoning under uncertainty is widely used in AI systems where:

- information is incomplete
- environments change dynamically
- decisions must be revised

These systems combine:

- symbolic logic
- uncertainty handling
- intelligent reasoning

2. Major Application Areas

Area	Usage
Expert Systems	Medical and technical diagnosis
Robotics	Dynamic environment handling
Natural Language Processing	Understanding ambiguous language
Autonomous Vehicles	Real-time decision making
Planning Systems	Updating goals and actions
Intelligent Agents	Adaptive reasoning

3. Expert Systems

Expert systems use:

- rules
- symbolic knowledge
- uncertainty reasoning

to imitate human experts.

Example

Medical Diagnosis System

Symptoms:

- Fever
- Headache
- Weakness

Initial AI conclusion:

- Viral infection

Later lab report:

- Malaria positive

Updated conclusion:

- Malaria

This uses:

- nonmonotonic reasoning
- belief revision
- TMS

4. Robotics

Robots work in changing environments.

They must:

- update beliefs
- revise decisions
- handle uncertainty

Example

Delivery Robot

Initial belief:

- Path is clear

New sensor input:

- Obstacle detected

Robot revises:

- Select alternate path

5. Natural Language Processing (NLP)

Human language is often:

- incomplete
- ambiguous
- context-dependent

Symbolic uncertainty reasoning helps NLP systems understand meaning.

Example

Sentence:

- "The bank is crowded."

Possible meanings:

- river bank
- financial bank

AI uses context to revise interpretation.

6. Autonomous Vehicles

Self-driving cars continuously receive:

- sensor data
- traffic updates
- obstacle information

They must revise decisions dynamically.

Example

Initial Decision

- Continue straight

New Information

- Accident ahead

Updated Decision

- Change route

Uses:

- TMS
- default reasoning
- nonmonotonic logic

7. Intelligent Personal Assistants

Assistants like:

- chatbots
- voice assistants
- recommendation systems

must manage changing user information.

Example

Initial assumption:

- User is at home

GPS update:

- User traveling

Assistant changes recommendations dynamically.

8. Planning Systems

Planning systems create and modify plans according to new conditions.

Example

Travel Planning

Initial plan:

- Flight available

New information:

- Flight cancelled

AI revises:

- Train reservation suggested

9. Cybersecurity Systems

Security systems use uncertain reasoning to detect threats.

Example

Initial belief:

- Login is normal

New evidence:

- Unusual location
- Multiple failed attempts

AI revises:

- Possible cyber attack

10. Financial Decision Systems

AI in finance handles:

- uncertain markets
- changing stock prices
- risk analysis

Example

Initial prediction:

- Stock price rising

New economic data:

- Market crash possibility

AI revises prediction dynamically.

11. Healthcare Systems

Healthcare AI systems continuously update diagnoses using:

- symptoms
- reports
- patient history

Example

Initial Symptoms

- Cough
- Fever

Initial diagnosis:

- Flu

New Test Result

- COVID positive

Updated diagnosis:

- COVID-19

12. Smart Home Systems

Smart systems use uncertain reasoning for automation.

Example

Initial assumption:

- Room empty

Motion detected:

- Lights turned ON

No motion for long time:

- Lights turned OFF

13. Case Study: Medical Expert System

Problem: Diagnosing diseases using uncertain symptoms.

AI Techniques Used

Technique	Purpose
Default Reasoning	Assume common diseases
Nonmonotonic Logic	Revise diagnosis

TMS	Maintain consistency
Bayesian Reasoning	Probability calculation

Workflow

1. Symptoms entered
2. Initial diagnosis generated
3. Tests performed
4. New evidence added
5. Diagnosis updated

Benefits

- Faster diagnosis
- Better decision support
- Dynamic reasoning

14. Case Study: Self-Driving Car

Problem: Dynamic road environment.

AI Challenges

- Traffic changes
- Pedestrians
- Roadblocks
- Weather uncertainty

Techniques Used

Technique	Role
Default Reasoning	Assume normal traffic
TMS	Update beliefs
Nonmonotonic Logic	Revise decisions
Symbolic Rules	Traffic law reasoning

Example

Initial belief:

- Green signal → move

New information:

- Pedestrian crossing

Updated action:

- Stop vehicle

15. Importance in Modern AI

Symbolic uncertainty reasoning is important because:

- real-world environments are uncertain
- information changes continuously
- AI must adapt dynamically

It provides:

- intelligent flexibility
- belief revision
- explainable reasoning

16. Future Scope

Future AI systems will increasingly use:

- hybrid symbolic AI
- explainable AI
- autonomous reasoning
- adaptive intelligent systems

Applications are growing in:

- healthcare
- transportation
- robotics
- cybersecurity
- smart cities